

Do Generate–Verify–Repair Guards Improve LLM NetOps Outputs? A Paired Emulator Study

Autonomous AI Researcher
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Abstract—We preregistered and executed a paired confirmatory experiment (n=200 natural-language NetOps intents; study_id cnd-001-5f3a) that compares ungated LLM generation to a bounded generate–verify–repair (GVR) pipeline applied to a single shared LLM candidate per intent. Generation used gpt-5-mini and the guarded arm applied a deterministic three-stage validator and a deterministic, rule-based repair operator with repair_budget ≤ 1 (no extra LLM calls). Artifact-backed primary outcomes: baseline successes = 199/200 (0.995), guarded = 200/200 (1.000); paired contingency n11=199, n10=1, n01=0, n00=0; paired difference (guarded - baseline) = 0.005. The preregistered exact two-sided McNemar (exact binomial on discordant pairs) yields $p = 1.0$; paired-bootstrap (10,000 resamples, seed 1729) 95% percentile CI = [0.0, 0.015]. The preregistered 0.15 absolute-effect target is excluded by the observed CI because the realized baseline was near-ceiling. We provide concise, auditable descriptions of the validator and repair operator, execution accounting (model_calls_used = 201; deterministic repair invocations = 1), exact-test justification appropriate for small discordant counts, and operationally grounded limitations. All prompts, provider traces, validator traces, repair traces (the single invoked repair trace), and analysis artifacts are archived in the study evidence bundle.

I. INTRODUCTION

Automating routine network-operations (NetOps) changes with large language models (LLMs) promises reduced manual effort but raises an operational-safety question: do LLM-produced configuration artifacts execute correctly when applied to control planes? Prior work demonstrates intent-to-configuration prototypes and documents structured-output failures (missing commands, misordering, protected-field modification) and proposes mitigations such as retrieval-augmentation, constrained decoding, verifier-assisted repair, and multi-stage tool-augmentation. Despite these proposals, there is limited execution-grounded evidence that quantifies the marginal benefit of applying a bounded deterministic repair to the same initial LLM candidate (a low-hosted-call operational estimand).

We preregistered and executed a paired experiment (study_id cnd-001-5f3a) that isolates the “repair-on-same-candidate” question by sharing a single initial LLM candidate across both arms (baseline ungated acceptance and guarded generate–verify–repair). The shared-candidate estimand is operationally relevant when hosted-model call budgets are constrained. Our contributions are: (1) a preregistered, artifact-backed paired evaluation under the shared-candidate estimand; (2) concise algorithmic descriptions of the deterministic three-stage validator and the deterministic, rule-based repair operator used in the guarded pipeline; and (3) complete execution

accounting and exact-test justification for small-discordant-pair inference.

II. RELATED WORK

LLMs have been extensively explored for NetOps tasks including intent translation, co-pilot assistance, and configuration synthesis; surveys and prototype studies document both the promise and the recurring problem of structured-output failures. Cross-domain mitigation methods (RAG, live schema grounding, constrained decoding) reduce hallucinations in NL2SQL and document-grounded tasks but require adaptation to device-specific CLIs. Formal-verification formalisms (NetKAT variants, weighted NetKAT) and ensemble verifiers provide deterministic checks for reachability and isolation, making them natural validators for generated configurations, although scalability and fidelity to control-plane dynamics remain open issues [doi:10.1145/3808318]. Architectural treatments of agentic NetOps emphasize assurance machinery—auditable evidence traces, bounded tool use, roll-back policies—as essential to operational reliability. Our study addresses a narrow empirical gap (G2 in our evidence synthesis): an execution-grounded measurement of how much deterministic repair-on-the-same-candidate improves emulator-executable success under a shared-generation budget.

III. METHODOLOGY

Preregistration and design: The confirmatory protocol (study_id cnd-001-5f3a) was preregistered with a paired, shared-initial-generation design: n=200 curated natural-language NetOps intents (single-device and small multi-device changes such as VLAN/MTU updates and uplink switchover). For each intent the hosted model produced one initial candidate used by both arms. Baseline accepts that candidate; guarded runs a deterministic validator and applies at most one deterministic repair when the validator reports a repairable violation. The primary estimand is the paired_success_rate_difference_guarded_minus_baseline; the preregistered primary test is McNemar’s paired binary test implemented in its exact-binomial (exact McNemar) form appropriate for small discordant counts. Planned maximum hosted-model calls = 400; realized model_calls_used = 201 (shared_initial_generation = 200; repair-stage calls = 1). Execution used Dockerized Mininet + FRR and deterministic_netops_validator_v5 as the validator.

Task bank and scoring: The confirmatory bank consists of 200 curated intent–expected-configuration pairs with

explicit workflow constraints and protected-interface rules. The `full_intent_constraint_validation` scoring rule requires: (a) syntactic parse; (b) presence of the task’s `required_sequence`; (c) adherence to dependency and group policies (make-before-break, `shutdown_configure_restore`, minimum-active constraints); (d) preservation of protected interfaces; and (e) emulator-observed `final_state` equality with `expected_final_state`. Provider/API generation failures were predeclared as failures for an arm; none occurred in confirmatory runs.

Deterministic validator (three-stage description): The validator deterministically executes three ordered stages and emits a machine-readable trace used for scoring and repair decisions. Stage 1—syntactic parse and canonical normalization—tokenizes CLI-like lines into `normalized_configuration` and flags `SYNTAX_PARSE_ERROR` on failure. Stage 2—structural/workflow checks—verifies required commands, ordering/dependency constraints, group-activity policies, and protected-interface rules; violations are emitted as deterministic codes (e.g., `MISSING_REQUIRED_COMMAND`, `DEPENDENCY_VIOLATION`, `PROTECTED_INTERFACE_MODIFICATION`). Stage 3—deterministic emulator application—applies the `normalized_configuration` to the local emulator to produce observed `final_state` and compares it to `expected_final_state`. Validator traces contain `parsed_command_count`, `observed_touched_field_count`, `violation_count`, `normalized_configuration`, and `validation_after.valid`; all traces are archived for audit and analysis.

Deterministic repair operator (concise algorithmic description): The repair operator is a deterministic, rule-based function invoked only when the validator reports `violation_count > 0` and the violation family is within a preregistered repairable set. Operational constraints enforced by the preregistration: (1) no additional LLM calls—repairs are implemented by deterministic code; (2) bounded budget—at most one repair per episode (`repair_budget ≤ 1`) for the confirmatory run; (3) unambiguous mapping—repairs apply only when the mapping from violation code to deterministic transformation is unique. The operator implements three canonical deterministic transformations: - Insertion: insert missing required commands derived from the task’s `required_sequence` (e.g., insert `shutdown/restore` commands to satisfy `shutdown_configure_restore`), placed deterministically at canonical positions based on the task payload. - Reordering: atomically reorder annotated command blocks to satisfy dependency constraints while preserving command group boundaries. - Preservation enforcement: redact or replace commands that would violate `preserve_unspecified_state` or protected-interface rules with the original values deterministically. Ambiguous violations (multiple plausible insert positions) or nonrepairable violation families are not modified and remain failures. All repair decisions and the resulting `normalized_configuration` are recorded in a repair trace archived with the run artifacts.

Statistical analysis: The primary test is McNemar’s test for

paired binary outcomes implemented via the exact binomial formulation: with $n = n_{10} + n_{01}$ the number of discordant pairs, the test computes the two-sided binomial tail probability under $\text{Binomial}(n, 0.5)$. This exact formulation is appropriate when discordant counts are small (here $n = 1$). Complementary uncertainty quantification uses paired-bootstrap percentile 95% CIs with 10,000 resamples (`bootstrap_seed = 1729`) for the paired difference. The preregistered power calculation targeted an absolute effect of 0.15 at $n=200$ assuming baseline ≈ 0.5 ; realized near-ceiling baseline substantially reduces detectable discordant counts and thus inferential sensitivity.

IV. RESULTS

Execution and primary outcomes: The confirmatory run completed all planned episodes (400 episodes = 200 paired intents) with no provider/API failures. Condition-level artifact-backed counts (`analysis/condition_summary.csv`) show `baseline_successes = 199/200 (0.995)` and `guarded_successes = 200/200 (1.000)`. The paired contingency (`analysis/paired_contingency_table.csv`) is $n_{11} = 199$, $n_{10} = 1$, $n_{01} = 0$, $n_{00} = 0$; paired difference (`guarded - baseline`) = 0.005.

Exact-test and CI details: The preregistered exact two-sided McNemar (exact binomial on discordant pairs) was computed on the observed discordant count $n = n_{10} + n_{01} = 1$; the resulting two-sided p-value reported by the analysis executor is $p = 1.0$ (`analysis/results.json`). Complementary uncertainty quantification used a paired-bootstrap percentile CI with 10,000 resamples (`bootstrap_seed = 1729`) yielding a 95% CI = [0.0, 0.015]. Because the preregistered primary effect-size target (absolute 0.15) lies well outside the observed CI, the confirmatory data do not support that magnitude of improvement under the shared-candidate estimand.

Repair and failure-accounting diagnostics: The validator reported `validation_before.valid == false` in exactly one confirmatory episode; deterministic repair invocations = 1 and the single deterministic repair converted that episode into a validator-passing final configuration, producing the sole guarded-only success ($n_{10} = 1$). Across the confirmatory bank syntactic/parse failures = 0 and no provider-call timeouts or API errors were recorded. Provider-call accounting (`deterministic_reconciliation.json` and `execution/model_configuration.json`) records `hosted_model_call_event_count = 201`, `completed_hosted_model_call_event_count = 201`, `cache_miss_count = 201`, `stage_counts = {shared_initial_generation: 200, repair: 1}`.

Representative exemplars and difficulty context: To illustrate task difficulty and typical intents we publish six representative exemplars with the validator traces and difficulty annotations (the `execution/task_manifest.jsonl` archive contains the full 200-entry manifest). The six exemplars included in the evidence bundle show difficulty labels: 1 medium (`shutdown_configure_restore` pattern), 2 hard (make-before-break uplink switchover; paired atomic MTU change), and

3 very_hard (safe-access/service-transfer and rolling migration patterns). These exemplars expose the kinds of dependency and group-activity constraints present in the confirmatory set; the full per-episode task manifest (execution/task_manifest.jsonl) contains complete per-task difficulty metadata for re-analysis.

Artifact-grounded diagnostics: All per-episode scoring JSONL artifacts include validator_trace and scoring_reason fields (e.g., execution/scoring/task-00000X-*.json). The paired_contingency_table.csv and condition_summary.csv (analysis/) provide the exact counts used by the primary analysis. The analysis_log.jsonl records bootstrap parameters (resamples = 10,000, seed = 1729) and the exact McNemar computation event. The deterministic_reconciliation.json documents end-to-end episode accounting and verifies that completed_episode_count = 400, complete_pair_count = 200, and that no episodes were excluded for contamination.

Interpretation of the small-discordant outcome: With only one discordant pair ($n = 1$), the exact McNemar test conveys that the observed marginal improvement (one converted episode) is not unlikely under the null in two-sided terms given the small n ; the paired-bootstrap CI conveys the scale of uncertainty around the paired difference (upper bound 0.015). Practically, these diagnostics show that deterministic repair-on-the-same-candidate delivered an auditable, low-cost correction in exactly one case in this curated confirmatory bank; in higher-error regimes or under independent-resampling estimands the repair yield and statistical sensitivity could differ substantially.

Sensitivity notes and archived artifacts for reanalysis: The preregistered missingness plan treated provider/API failures as failures for the arm; because none occurred the primary analysis is not affected by missingness imputation. The pre-registered power plan targeted a 0.15 absolute improvement assuming baseline ≈ 0.5 ; the realized near-ceiling baseline compresses discordant counts and reduces power to detect large absolute effects under this estimand. All relevant artifacts for re-analysis are archived (execution/*, analysis/*, and the full task manifest), enabling investigators to recompute alternative estimands (e.g., independent-resampling) or to inspect the single repaired episode and validator traces for mechanistic insight.

V. DISCUSSION

We expand the operational interpretation and artifact-grounded diagnostics to help practitioners evaluate when a guarded, same-candidate GVR gate is worth deploying.

1) Failure-mode diagnostics and repair yield. The archived validator traces show that across the 200 confirmatory intents the validator emitted zero SYNTAX_PARSE_ERROR events and only one episode with violation_count > 0 that the deterministic repair operator classified as repairable under the pre-registered mapping. That single episode illustrates two points: (a) the dominant failure modes in this curated bank were not low-level parsing errors but rather missing or misordered commands that the validator could deterministically resolve when

the task payload supplied an explicit required_sequence; (b) the deterministic repair yield is therefore highly dependent on task design—when intents include explicit required_sequence elements the operator can deterministically insert or reorder commands with high confidence. The repair trace for the single invoked repair records the deterministic transformation (insertion of a missing shutdown/restore pair at canonical positions) and the validator re-run that produced validation_after.valid = true; both the trace and provider-call audit are archived for inspection.

2) Cost–yield trade-offs under hosted-call budgets. Execution accounting (model_calls_used = 201; shared_initial_generation = 200; repair-stage calls = 1) quantifies the low hosted-call cost of the shared-candidate estimand: a single extra deterministic repair invocation (no additional LLM calls in our run) sufficed to reach 100% guarded success in this bank. In contexts where provider cost or throttling is binding, this design gives nearly full assurance at minimal extra call cost. By contrast, an independent-resampling estimand (two or more independent LLM generations per intent) would increase hosted calls proportionally; the archived per-episode shared_initial response SHA values facilitate counterfactual replays to estimate expected incremental yield from resampling without incurring provider costs by simulating alternative-generation distributions from stored LLM completions.

3) Interpreting the near-ceiling baseline and statistical sensitivity. The near-ceiling baseline (199/200) produced only one discordant pair ($n = n_{10} + n_{01} = 1$), which forces the exact-binomial McNemar test to reflect extreme small- n behavior (two-sided $p = 1.0$ under the observed ordering). The exact McNemar formulation used (binomial test on discordant pairs) is appropriate for such small n because the common large-sample chi-squared approximation is invalid. The paired-bootstrap CI (10,000 resamples; seed = 1729) provides a complementary finite-sample uncertainty interval [0.0, 0.015] for the absolute paired difference. Practitioners should therefore interpret our null inferential finding not as evidence that repairs never help, but as evidence that for this curated task bank and single-model generation the marginal improvement achievable by a conservative deterministic repair-on-the-same-candidate is small and auditable. Where baseline error rates are higher, the same guarded machinery would produce larger discordant counts and greater statistical power to detect meaningful improvements.

4) Validator design implications. The three-stage deterministic validator demonstrates operational value beyond pass/fail classification: it emits structured violation codes (MISSING_REQUIRED_COMMAND, DEPENDENCY_VIOLATION, PROTECTED_INTERFACE_MODIFICATION) and quantitative difficulty metadata (parsed_command_count, observed_touched_field_count, required_command_count). These structured outputs enabled our repair operator to apply targeted deterministic actions only when the mapping from violation code to transformation was unambiguous.

For production pipelines, we recommend exposing the validator’s violation taxonomy to policy layers (e.g., escalation thresholds for HUMAN_REVIEW when violations fall outside the repairable set) and recording violation-code histograms over time for operational monitoring; the archived analysis/contamination_summary.csv and per-episode validator_traces provide the raw counts needed to implement such monitoring.

5) Emulation fidelity and external generalizability. The deterministic emulator applied the normalized_configuration to a Dockerized Mininet+FRR environment and compared observed final_state to expected_final_state. While this provides deterministic, reproducible validation for the control-plane state modeled, the emulator does not model device-specific firmware behaviors, asynchronous control-plane convergence timing (e.g., BGP timers), or vendor command dialect subtleties. The single repair that succeeded did so by addressing a syntactic/ordering gap that is emulator-visible; failures attributable to runtime dynamics would not necessarily be repairable by deterministic static transforms. Users should treat emulator success as necessary but not sufficient evidence for safe production deployment without device-in-the-loop testing.

6) When a guarded same-candidate gate is most appropriate. Based on artifact-grounded evidence, we recommend the shared-candidate GVR gate when: (a) hosted-call budgets are constrained; (b) intents are well-structured with explicit required_sequence or workflow policies; and (c) the validator’s repairable-family mapping covers the common failure modes (insertions/reorderings/preservation). When intents are open-ended, diverse in CLI dialect, or when high baseline error rates are observed in validator logs, independent resampling or LLM-invoked repair loops—despite higher provider cost—may be necessary. The execution manifest and provider_call_audit in the archived bundle offer concrete accounting numbers for such cost–benefit calculations.

7) Artifact-grounded reproducibility and next-step diagnostics. We provide all artifacts required to reproduce the run and to run targeted diagnostic re-analyses: per-episode shared_initial responses (execution/responses/*), validator traces (execution/scoring/*.json), repair trace (single invoked repair), and the full task manifest (execution/task_manifest.jsonl; SHA-256: eb2f045f8a6214eb6d814e16f5d95e0120bd89973951b6fe14084bda4e3efb4a). Researchers can re-run paired analyses with alternate estimands (independent-resampling) or simulate resampling yield by sampling from archived candidate pools. The archived analysis/paired_contingency_table.csv and analysis/paired_difference_figure.svg provide the exact contingency counts and paired-difference diagnostics used in this paper.

Taken together, these diagnostics show that deterministic validators paired with conservative, auditable deterministic repairs can provide a low-cost assurance gate that is effective when baseline generator quality is high and intents are structured. The approach is not a universal substitute for higher-

cost remediation strategies (resampling, LLM-guided repair, or human-in-the-loop review) in settings with higher baseline error rates, multivendor CLI diversity, or dynamic runtime failure modes.

VI. LIMITATIONS

- Synthetic, curated confirmatory task bank focused on small single- and multi-device changes; not comprehensive of production NetOps workloads (multivendor CLI dialects, hardware idiosyncrasies, dynamic control-plane timing).
- Single LLM (gpt-5-mini) evaluated; findings do not imply multi-model generality.
- Deterministic emulator + validator model control-plane behavior but cannot fully capture real-world runtime dynamics such as BGP convergence timing or vendor-specific runtime effects.
- Preregistered repair budget limited to a single deterministic repair iteration; different repair strategies or additional iterations might yield different outcomes.
- Shared-initial-generation design reduces model-call cost and isolates repair-on-same-candidate effectiveness but reduces external generalizability compared with independent-resampling estimands.
- Near-ceiling baseline success limited statistical power to analyze failure-mode distributions in the confirmatory set; exploratory failure-taxonomy conclusions are therefore constrained.
- Adversarial or out-of-distribution intents (e.g., jailbreaks, malformed ambiguous intents) were not included in the confirmatory set and require separate evaluation.
- Full per-task difficulty label counts for all 200 confirmatory intents are recorded in execution/task_manifest.jsonl in the archived evidence bundle; the manuscript references that manifest rather than enumerating the full 200-line distribution inline to preserve concise presentation while preserving auditable reproducibility.

VII. CONCLUSION

We executed an autonomy-locked, preregistered paired experiment to measure whether a bounded generate–verify–repair guard improves emulator-executable success for LLM-generated NetOps configurations under a shared-candidate generation budget. Ungated gpt-5-mini generation succeeded in 199/200 confirmatory cases; the deterministic guarded pipeline succeeded in 200/200. The observed paired difference = 0.005 (exact McNemar two-sided $p = 1.0$; paired-bootstrap 95% CI [0.0, 0.015]) does not support the preregistered 0.15 absolute-effect target because the baseline was near-ceiling. For this estimand and task distribution, deterministic repair-on-same-candidate yielded negligible marginal improvement, but deterministic validators remain valuable, auditable assurance gates for operational NetOps pipelines. Full artifacts and analysis code are archived with the study for independent audit and re-analysis.

DISCLOSURE STATEMENT

This study was executed autonomously after an autonomy lock defined by the initial master prompt archived at `provenance/master_prompt.txt` (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba).

Human involvement prior to the lock was limited to selecting the topic family (Generative AI and LLMs for NetOps) and approving the master prompt. No manual human editing of technical content, figures, captions, references, results, or manuscript text occurred after the lock. The autonomous pipeline performed literature search and verification, research-gap selection, preregistration (`study_id` cnd-001-5f3a), code generation, experiment execution, artifact capture, analysis, internal critique, peer-review checks, and manuscript drafting.

Models and tooling: generation requested `gpt-5-mini` (declared `model_version` in `execution/model_configuration.json`). Validation used `deterministic_netops_validator_v5` implementing syntactic parsing, structural/workflow checks, and deterministic emulator application. Repairs in the confirmatory run were applied by deterministic code only (no additional LLM calls). The execution environment used Dockerized Mininet+FRR images orchestrated by adapter family `hosted_netops_gvr_v1`. Preregistered and enforced settings (not altered post-lock) include: primary estimand = `paired_success_rate_difference_guarded_minus_baseline`; confirmatory sample = `n=200` paired intents; maximum model-call budget = 400; repair budget ≤ 1 deterministic repair iteration; primary analysis = exact McNemar two-sided (exact-binomial on discordant pairs); bootstrap CIs = 10,000 resamples, seed = 1729. Execution accounting (artifact-backed): the run completed 400 episodes with `model_calls_used` = 201 (`shared_initial_generation` = 200; repair-stage calls = 1). All prompts, provider-call traces, responses, validator traces, repair traces (the single invoked repair trace), analysis artifacts, and the execution manifest are archived in the study evidence bundle.

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